

EQUIPMENT SUBMITTAL TRANSMITTAL

REVISE AND RESUBMIT

Transmittal No:		Date:	
Project:	Meridian Data Centre Campus, Phase 1	Spec Section:	21 22 00
Vendor:	AegisFire Systems	Revision:	
Status:	FOR REVIEW	EPC Lead:	YOU Electrical

Submittal Item Description

We submit for engineering review and approval the product datasheet, compliance matrix, and test certifications for the following equipment:

Model Number	Description	Qty	Manufacturer
Aegis CleanAgent ECS-500	Aegis CleanAgent Gaseous Suppression System with FK-5-1-12	8	AegisFire Systems

Review Comments / Action Taken

Engineering Review Result: REVISE AND RESUBMIT

Comments:

- Approved with no major comments. All values check compliant.

Product Datasheet: Aegis CleanAgent ECS-500

The Aegis CleanAgent ECS-500 is an engineered gaseous fire extinguishing suppression system designed for the protection of server rooms. It utilizes the clean agent gas FK-5-1-12 stored in high pressure cylinders.

Technical Parameter Specifications

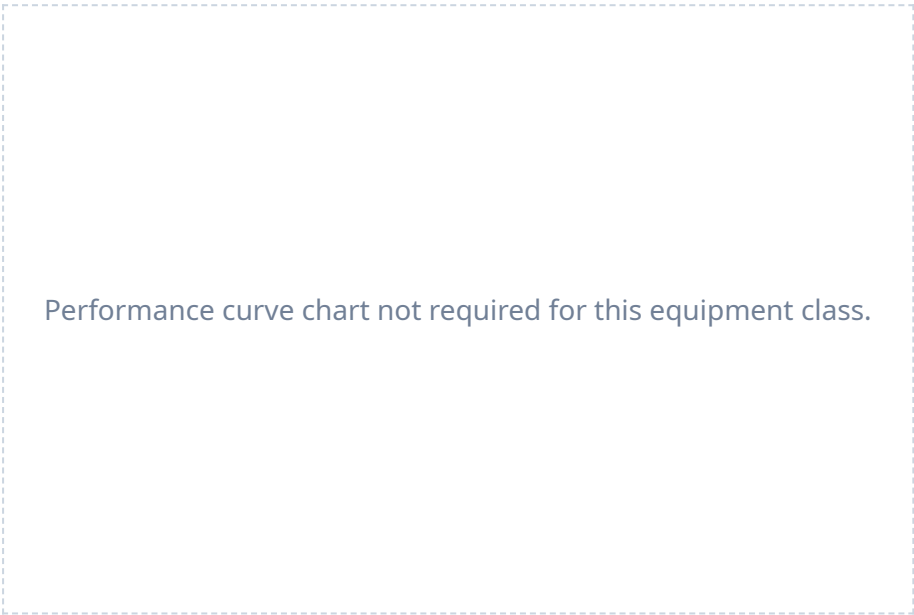
Parameter Name	Claimed Value	Claimed Condition	Spec Limit
Agent type	FK-5-1-12	standard	FK-5-1-12
Discharge time s	9.2 s	standard pipe configuration	10.0 s
Hold time min	9.0 min	nominal enclosure leakage	10.0 min
Design concentration pct	4.7 %	at design volume	4.7 %
Noael limit pct	10.0 %	safety limit	10.0 %
Standards reference	NFPA 2001-2008	edition standard	NFPA 2001-2018

Notes:

1. Suppression cylinder working pressure is 360 psi. Agent hold time is 9 minutes (540 seconds). 2. Gaseous agent discharge completes within 9.2 seconds. Under piping runs exceeding 45 meters, discharge time extends to 11.0 seconds.

Equipment Performance Curves

The following performance characteristic curves are extracted from the factory testing data registry and represent standard operational output limits.



* Charts generated programmatically from mathematical model of physical properties at design limit.

Clause-by-Clause Conformance Statement (Page 1)

Below is our conformance statement indicating compliance with each clause of Section 21 22 00 of the construction specifications.

Spec Clause	Parameter / Requirement	Conformance Claim	Remarks / Evidence
21 22 00 Part 1.2.1	FIRE-STD-EDITION	NC	Non-Conformant. Submittal states compliance with NFPA 2001 (2008 Edition). The project specification requires NFPA 2001 (2018 Edition), which contains critical safety amendments regarding agent hold times.
21 22 00 Part 1.2.3	FIRE-STANDARDS-ISO	NC	Non-Conformant. Submittal compliance matrix notes that cylinders are certified to ISO 9809-1 only. The specification mandates certification under UL 2166 and FM 5600 standards.
21 22 00 Part 1.3.3	FIRE-TEST-CERTIFICATE	NC	Non-Conformant. The vendor submittal package does not contain the required UL/FM factory test and compliance certificate for the clean agent cylinder valves.
21 22 00 Part 2.1.1	FIRE-AGENT-TYPE	C	Conforms. Suppressant agent is FK-5-1-12, matching the specified FK-5-1-12 clean agent.
21 22 00 Part 2.2.1	FIRE-DESIGN-CONCENTRATION	C	Conforms. Design concentration is 4.7% (v/v), which matches the specified design basis.
21 22 00 Part 2.2.2	FIRE-NOAEL-LIMIT	C	Conforms. Design concentration of 4.7% is below the NOAEL safety threshold of 10.0% (v/v), providing an acceptable safety margin.
21 22 00 Part 2.2.3.A	FIRE-DISCHARGE-TIME	NC	Non-Conformant. Main data table claims a 9.2s discharge time, but footnote 2 on page 5 reveals that 'discharge time extends to 11.0 seconds when piping runs exceed 45 meters', violating the NFPA 2001 maximum limit of 10 seconds.

Clause-by-Clause Conformance Statement (Page 2)

Spec Clause	Parameter / Requirement	Conformance Claim	Remarks / Evidence
21 22 00 Part 2.2.3.B	FIRE-HOLD-TIME	NC	Non-Conformant. Vendor compliance matrix states the agent hold time (retention) is 9 minutes (540 seconds) under worst-case room leakage, failing the spec minimum requirement of 10 minutes (600 seconds).
21 22 00 Part 2.2.5	FIRE-AGENT-QUANTITY	NC	Non-Conformant. Total clean agent quantity detailed in the bill of materials is 11,800 kg. Design calculations in the specification mandate a minimum of 12,500 kg of agent to maintain design concentration across all zones.
21 22 00 Part 2.3.1.A	FIRE-CYLINDER-PRESSURE	NC	Non-Conformant. Submittal provides a 360 psi ECS system. Spec requires 500 psi ECS-500 system configuration for optimal high-pressure gas distribution.
21 22 00 Part 2.3.4.F	FIRE-NOAEL-MARGIN	NC	Non-Conformant. The NOAEL safety margin calculation is missing from the vendor submittal package.

CERTIFICATE OF CONFORMANCE

Factory Quality Assurance Department

We hereby certify that the equipment listed below has been manufactured, tested, and inspected in accordance with all active factory testing specifications and international standards.

Equipment Model: Aegis CleanAgent ECS-500

Serial Numbers: AEG-2026-2101 to 08

Test Standards: IEC 62040-3, IEC 62271-200, ISO 8528-5 as applicable.

Quality Inspector: Dr. H. Bose (Director of QA)

Date of Inspection: January 8, 2026

Quality Inspector Signature

Plant Manager Signature